

Neural Mental Maps

Exploration Strategies for INR-based World Models

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Abstract

Autonomous exploration in unknown environments requires agents to carefully balance immediate information gain with long-term navigation goals. While active learning via Implicit Neural Representations (INRs) has proven highly effective for continuous world modeling, its performance under strict physical movement budgets remains underexplored. We address this by benchmarking constrained INR-based acquisition functions against classical geometric heuristics (Frontier) and reinforcement learning (PPO). Beyond standard reconstruction metrics (PSNR, MSE), the learned world models are evaluated on downstream tasks like path planning and collision avoidance, highlighting the critical trade-offs between greedy uncertainty-driven active learning and long-horizon RL strategies in realistic embodied scenarios.

1. Introduction

Exploration is a fundamental challenge in autonomous robotics. Traditional active learning strategies efficiently map unknown environments, but they frequently assume unconstrained observation capabilities (which can lead to teleportations). In realistic embodied scenarios, strict movement budgets create two main challenges: agents can only query local areas, and short-sighted movements often trap them in suboptimal regions.

While Implicit Neural Representations (INRs) have emerged as a great standard for continuous world modeling, their exploration strategies are typically evaluated without physical constraints. We aim to overcome this by evaluating whether active INR-based world models can adapt to physical constraints to provide superior exploration compared to established robotics baselines.

Specifically, we address the following research questions:

- **RQ1:** Do INR-based active strategies (designed for unconstrained settings) still outperform heuristic exploration methods specifically built for the embodied case when subjected to reachability constraints?
- **RQ2:** Can RL-trained policies, which learn long-horizon strategies to overcome myopia, outperform both uncertainty driven heuristics and classical geometric methods?

To answer these questions, we implemented constrained acquisition functions to compare principled active strategies, classical robotics baselines, and end-to-end RL policies.

Our empirical analysis focuses on how standard active learning degrades under reachability constraints. Furthermore, we validate the exploration viability not just via abstract reconstruction metrics (NRMSE, PSNR), but by evaluating the learned models on practical downstream path planning and collision avoidance tasks.

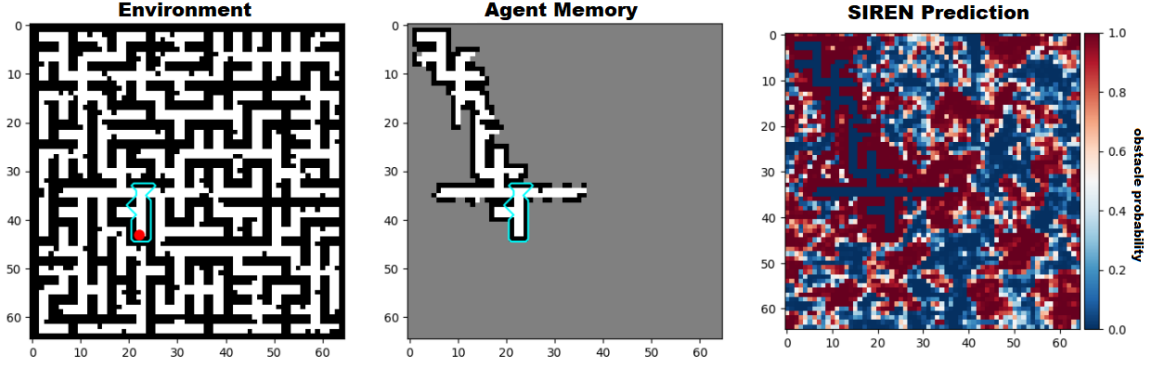


Figure 1: The environment in which agent has to navigate (left). An image of fields already seen in previous steps and observed at the moment (middle). The SIREN model representing the world (right), obtained using limited information about environment. The edge of region observed from current position (red dot) is marked down by light-blue line.

2. Related Work

Implicit neural representations and active learning. INRs like SIREN [8] excel at continuous signal modeling. Despite its efficiency in targeting epistemic uncertainty, standard active learning typically assumes unrestricted movement capabilities. We investigate how these methods degrade under strict physical budgets.

Embodied exploration and robotics. Geometric strategies like frontier exploration [10] expand map coverage by continuously targeting the boundaries of known space. However, these methods are inherently myopic and lack environment-specific priors.

Reinforcement learning for exploration. To overcome classical limitations, deep RL algorithms (like PPO [7] or SAC [2]) approach exploration sequentially. By learning long-horizon policies, they successfully avoid the local optima that frequently trap acquisition functions.

3. Experiments

SIREN Architecture and Training Setup. We formulated the continuous world modeling task as learning an implicit function $f_\theta : \mathbb{R}^2 \rightarrow [0, 1]$ mapping spatial coordinates to occupancy or traversal cost [5]. We utilized the SIREN architecture [8] consisting of 3 hidden layers with 256 neurons each, utilizing a linear transformation at the final output. Weights were initialized using architecture-specific frequencies ($\omega_0 = 30.0$ for the first layer and $\omega_0 = 10.0$ for the rest). During each exploration step, the model is updated on the

accumulated memory using the Adam optimizer (learning rate 5×10^{-4}) for 200 epochs. To calculate model uncertainty, we implemented Monte Carlo Dropout (rate = 0.1) [1; 9] exclusively on the penultimate layer. During map evaluation, network weights are frozen.

Environments and Agent Perception. We evaluated the agents across five diverse 2D environments. The first, **Discrete Mazes** (both *Standard* one-cell and *Wide* three-cell variants), are generated via Prim’s algorithm [6] and feature hard walls. The second, **Continuous SDF**, uses a distance map representing smooth circular obstacles (1.0 for collisions, 0.0 for safety). The final three environments **Radiation Mapping**, **Perlin Terrain**, and **Gaussian Random Field** are obstacle-free continuous cost maps. Generated using Gaussian sources, filtered procedural noise, and smoothed random fields respectively. In all environments, the agent operates under partial observability, updating the INR via raycasting every 1° up to a vision radius of 16 for mazes and 10 for the others, (see Figure 2 for a visualization).

Exploration Baselines. To benchmark exploration performance, the maximum Euclidean distance for a single step was strictly limited to 8 across all environments. We compared four SIREN-based active strategies: **MaxVar** (greedy variance maximization), **BatchDPP** (Determinantal Point Process for spatially diverse batch selection [3]), **InfoGain RRT** (stochastic planning utilizing variance [4]), and **Frontier** [10] (geometric boundary exploration). These were contrasted with a RL baseline, **PPO** [7].

Evaluation Metrics. We evaluate exploration performance based on both world model reconstruction quality and its viability for downstream tasks. For reconstruction quality, we compute the Mean Squared Error (**MSE**) between the predicted field and the ground-truth. Depending on the environment’s physics, we report either the Peak Signal-to-Noise Ratio (**PSNR**) for normalized occupancy, or the Normalized Root MSE (**NRMSE**) for continuous fields. To measure downstream task viability, we evaluate 200 path-planning queries. Paths are planned on the agent’s reconstructed map and validated against the ground-truth environment. We report three metrics: (1) **Success Rate**, the fraction of planned paths that reach the goal collision-free; (2) **Collision Rate**, the fraction of paths that intersect physical obstacles; and (3) **Path Cost Ratio**, the average cost of successful paths divided by the optimal path cost.

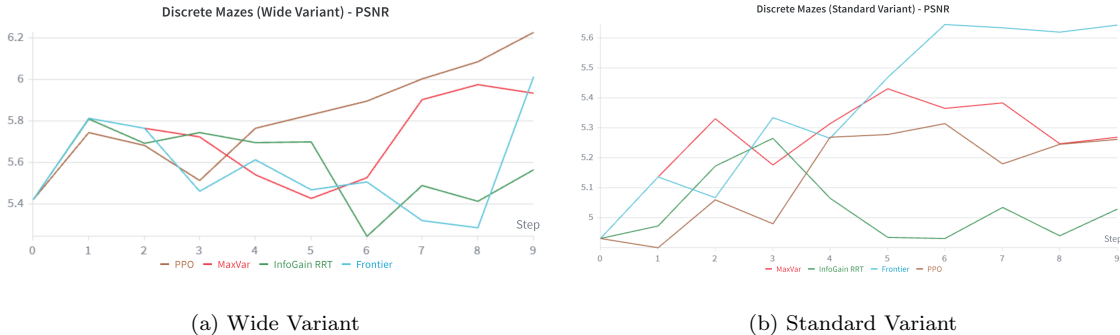


Figure 2: Comparison of exploration performance (PSNR vs. Exploration Steps) in Discrete Maze environments under regression mode. (a) In the Wide variant featuring stochastic obstacles, long-horizon strategies (PPO Baseline) consistently maintain superior reconstruction quality by avoiding dead-ends. (b) In the Standard variant with narrow corridors, myopic active learning heuristics (like MaxVar and Frontier) degrade significantly, repeatedly getting trapped in local optima.

4. Results

Table 1 summarizes downstream task viability. While variance-maximizing heuristics map smooth topologies efficiently ($> 90\%$ Success Rate), strict reachability constraints expose their myopia. In Wide Mazes, they frequently hit obstacles ($> 30\%$ Collision Rate).

The PPO baseline overcomes this. Maintaining competitive Success Rates in Wide Mazes (14.7% vs 15.7% for MaxVar), PPO halves the Collision Rate (14.0%). In restricted Standard Mazes (Appendix 6.1), despite a 10-step horizon limiting all methods, PPO consistently minimizes collisions. This confirms RL learns robust policies prioritizing safe navigation over purely uncertainty-driven exploration.

Method	Smooth Env.		Wide Mazes	
	Succ. $\uparrow$	Coll. $\downarrow$	Succ. $\uparrow$	Coll. $\downarrow$
MaxVar	90.3%	9.7%	15.7%	30.7%
BatchDPP	90.4%	9.6%	15.7%	30.6%
InfoGain RRT	90.6%	9.4%	9.3%	4.7%
Frontier	90.3%	9.8%	15.5%	8.7%
PPO	90.3%	9.8%	14.7%	14.0%

Table 1: Success and Collision Rates (Averaged over 3 seeds). Smooth environments aggregate GRF, Perlin, Radiation, and SDF. Full results, including Standard Mazes and Path Costs, are detailed in Appendix 6.1.

5. Conclusions

In this study, we investigated the performance of active learning strategies driven by Implicit Neural Representations (INRs) under strict physical movement budgets. Our evaluation reveals a critical dichotomy in exploration performance depending on the environment’s topology.

Addressing RQ1, our findings indicate that in smooth, open environments (such as the Gaussian Random Field and Continuous SDF), variance-maximizing active learning (MaxVar) significantly outperforms classical heuristics. The capacity of INRs to estimate epistemic uncertainty allows the agent to highly efficiently map continuous structures. Furthermore, across all tested environments, modeling the INR output via continuous regression systematically yielded superior and more stable reconstructions (higher PSNR, lower MSE) compared to discrete classification modes.

However, the introduction of reachability constraints and hard obstacles (Discrete Mazes) exposes the inherent myopia of standard active acquisition functions. Both uncertainty-driven methods (MaxVar) and classical geometric heuristics (Frontier) suffer from severe temporal credit assignment issues, frequently becoming trapped in dead-ends.

Addressing RQ2, we demonstrate that end-to-end reinforcement learning policies successfully overcome these limitations. The PPO Baseline, while exhibiting higher variance across different initialization seeds, effectively learns robust, long-horizon exploration strategies. It consistently avoids the local optima that cripple greedy algorithms, proving that RL is a highly viable alternative for embodied exploration where movement budgets and physical constraints cannot be ignored.

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6. Appendix

6.1 Detailed Downstream Task Results

Table 2 provides the complete breakdown of downstream task performance across all evaluated environments and constraints, including the Path Cost Ratio and performance in the highly constrained Standard Discrete Mazes. In Standard Mazes, the strict 10-step horizon prevented PPO from completing the path successfully across some seeds, resulting in a Not Applicable (N/A) Path Cost Ratio.

Table 2: Detailed downstream task performance broken down by environment constraints (Averaged over 3 seeds). Smooth Environments aggregate GRF, Perlin, Radiation, and SDF. Bold values indicate the best performance. It is important to mention that for the *Standard* maze, only a small part of the whole environment was discovered; therefore, path planning on the learned representation was a much more difficult task.

Method	Success Rate $\uparrow$	Collision Rate $\downarrow$	Path Cost Ratio $\downarrow$
<i>Continuous & Smooth Environments</i>			
MaxVar	90.3%	9.7%	1.013
BatchDPP	90.4%	9.6%	1.014
InfoGain RRT	90.6%	9.4%	1.014
Frontier	90.3%	9.8%	1.015
PPO	90.3%	9.8%	1.015
<i>Wide Discrete Mazes (Moderate Constraints)</i>			
MaxVar	15.7%	30.7%	1.003
BatchDPP	15.7%	30.6%	1.003
InfoGain RRT	9.3%	4.7%	1.000
Frontier	15.5%	8.7%	1.000
PPO	14.7%	14.0%	1.000
<i>Standard Discrete Mazes (Severe Constraints)</i>			
MaxVar	2.7%	0.0%	1.000
BatchDPP	2.5%	0.0%	1.000
InfoGain RRT	1.5%	0.0%	1.000
Frontier	2.8%	0.0%	1.000
PPO	0.7%	0.0%	1.000